

The Efficacy of Paroxetine and Placebo in Treating Anxiety and Depression: A Meta-Analysis of Change on the Hamilton Rating Scales

Background Previous meta-analyses of published and unpublished trials indicate that antidepressants provide modest benefits compared to placebo in the treatment of depression; some have argued that these benefits are not clinically significant. However, these meta-analyses were based only on trials submitted for the initial FDA approval of the medication and were limited to those aimed at treating depression. Here, for the first time, we assess the efficacy of a selective serotonin reuptake inhibitor (SSRI) in the treatment of both anxiety and depression, using a complete data set of all published and unpublished trials sponsored by the manufacturer.

Methods and Findings GlaxoSmithKline has been required to post the results for all sponsored clinical trials online, providing an opportunity to assess the efficacy of an SSRI (paroxetine) with a complete data set of all trials conducted. We examined the data from all placebo-controlled, double-blind trials of paroxetine that included change scores on the Hamilton Rating Scale for Anxiety (HRSA) and/or the Hamilton Rating Scale for Depression (HRSD). For the treatment of anxiety ($k = 12$), the efficacy difference between paroxetine and placebo was modest ($d = 0.27$), and independent of baseline severity of anxiety. Overall change in placebo-treated individuals replicated 79% of the magnitude of paroxetine response. Efficacy was superior for the treatment of panic disorder ($d = 0.36$) than for generalized anxiety disorder ($d = 0.20$). Published trials showed significantly larger drug-placebo differences than unpublished trials (d 's = 0.32 and 0.17, respectively). In depression trials ($k = 27$), the benefit of paroxetine over placebo was consistent with previous meta-analyses of antidepressant efficacy ($d = 0.32$).

Conclusions The available empirical evidence indicates that paroxetine provides only a modest advantage over placebo in treatment of anxiety and depression. Treatment implications are discussed.